Development & Integration

Date	2 October 2026
Team ID	6abca5b16f395c49751f124d
Project Name	LegalEase: AI-Powered Legal Document Generator
Maximum Marks	Project Deliverable

Frontend Development

Streamlit provides the LegalEase user interface with document inputs, a Generate Document action, generated-document preview, editing control and TXT/DOCX/PDF downloads.

Backend Logic Implementation

FastAPI exposes service endpoints such as /, /health and POST /generate. Pydantic validates the request before generation. The backend remains separate from the frontend so Streamlit communicates with Gemini through FastAPI rather than calling the model directly.

Gemini API Integration

A dedicated Gemini generator/service initializes the current Google GenAI client using GEMINI_API_KEY and a configurable GEMINI_MODEL. It constructs the prompt, sends the request, checks the response and returns generated text to the API layer.

Export Integration

- TXT: latest edited content encoded as UTF-8.
- DOCX: professional headings, margins, paragraphs, footer and optional LegalEase branding.
- PDF: wrapped multi-page content with margins, headings, footer and page handling.

GitHub Integration

- .env excluded using .gitignore.
- .env.example contains placeholders only.
- requirements.txt contains project dependencies.
- README documents setup, architecture, usage, testing and disclaimer.